INCH-POUND

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30 October 1979

MILITARY SPECIFICATION

REGULATOR, VOLTAGE: 10548988

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 33-volt regulator, part of converter assembly 10549013 used in Power Supply Subassembly PP6324\TPM-22, part of Radar Test Set AIJ/TPM-22, and in Power Supply Subassembly PP4812/VPS-2, park of Radar Set ANiVPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, General Specification Governing the Manufacture and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
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MIL-I-45607- Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-sTD-lo9- Quality Assurance **Terms and Definitions**
MIL-sTD-810- **Environmental Test Methods and** Engineering Guidelines
MIL-sTD-2000- Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from Military Specifications and Standards, Bldg. AD, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

2.1.2 Other Government documents, drawings, and publicationso The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein **Unless otherwise specified, the issues shall be those in effect on the** date of the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548988 33-Volt Regulator

PACKAGING DATA SHEET

PS10548988 - Packaging of 33-Volt Regulator

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has **been obtained.**

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

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10548988,all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the 33-volt regulator shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
Part identification and marking
:: Workmanship

3.3 Environmental service conditions.

3.3.1 Vibration. The 33-volt regulator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 40 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
@ The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-40-10 hertz sweep time is 2 minutes 30 seconds ± 10%.

3.3.2 Shock. The 33-volt regulator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The 33-volt regulator shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The 33-volt regulator shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The 33-volt regulator shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 degrees F and +155 degrees F.

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3.3.4.2 Non-operating (storage) . The 33-volt regulator shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization) , and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating) The 33-volt regulator shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess "of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the 33-volt regulator.

3.5 Performance. The 33-volt regulator shall produce results as given in 3.5.1 and 3.5.2 when supplied with the following:

44.5 ± 8.5 volts direct current (Vdc), regulated to ± 0.5 percent between P1-6 (+)1/, P1-8 (+), P1-7 (-), and P1-9 (-).

b. 22 ± 1 kilohertz (KHz), 71.5 ± 12.5 volt (V) peak-to-peak **square wave between P1-1 and P1-2 and between PI-3 and P1-4.**

3.5.1 Output terminals E71-E72.

3.5.1.1 Output voltage level potentiometer R2. R2 **shall control the level of output voltage between E71 (+) and E72 (-). The nominal output voltage shall be 33.00 ± 0.10 Vdc with a load current of 0.75 ± 0.2 ampere direct current (Adc).**

3.5.1.2 Current limit Potentiometer R17. R17 shall establish a point at which the current limiting circuitry **becomes effective. with a short applied between E71 and E72, the maximum current drain shall be controlled by R17 and associated circuitry and shall not exceed 1.6 Adc. The output shall automatically recover after removal of the short circuit.**

3.5.1.3 Voltage regulation line. The output voltage shall not vary more than ± 0.33 V with an input voltage variation from 36 to 53 Vdc and a constant load current of 0.75 Adc.

3.5.1.4 Voltage regulation, load. with a constant input voltage of 44.5 ± 8.5 Vdc, the output voltage shall not vary more than ~ 0.33 V when the output current is varied from O to 0.7S Adc.

3.5.1.5 Output ripple. The ripple shall not exceed 0.33 V peak-to-peak when the input voltage is 44.5 ± 0.44 Vdc and the load

The information in parentheses refers to the polarity applied to this terminal. This notation Shall be applied throughout this specification.

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current is 0.75 Adc. Exclude high-frequency spikes which occur at approximately 25 usec. intervals.

3.5.2 Output terminals E73 - E74.

3.5.2.1 Output voltage level potentiometer R25. R25 shall control the level of output voltage between E73 (+) and E74 (-). The nominal output voltage shall be 33.00 ~ 0.10 Vdc with a load current of 0.75 Adc.

3.5.2.2 Current limit potentiometer R40. R40 shall establish a point at which the current limiting circuitry becomes effective. With a short applied between E73 and E74, the maximum current drain shall be controlled by R40 and associated circuitry and shall IUJ-L exceed 1.6 Adc. The output shall automatically recover after removal of the short circuit.

3.5.2.3 Voltage regulation, line. The output voltage shall not vary more than ± 0.33 V with an input voltage variation from 36 to 53 Vdc and a constant load current of 0.75 Adc.

3,5.2.4 Voltage regulation, load. With a constant input voltage of 44.5 ± 8.5 Vdc, than ± 0.33 V when the output current is varied from O to 0.75 Adc.

3.5.2.5 Output ripple. The ripple shall not exceed 0.33 V peak-to-peak when the input voltage is 44.5 ± 0.44 Vdc and the load current is 0.75 Adc. Exclude high-frequency spikes which occur at approximately 25 psec. intervals"

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3) .

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-sTD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define guality assurance terms used herein.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The requirement for first article and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract and the applicable drawing. The sample for the first article tests shall consist of three 33-volt regulators manufactured in the same manner, using the same materials, equipment, processes and procedure as in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.3.1.1 Government testing. When the Government is responsible for conducting first article tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and the applicable drawings. The contractor shall submit a record of this inspection with the sample including certified test report (CTR) or certificate of conformance (COC) for the materials.

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4.3.1.2 Contractor testing. When the Contractor is responsible for conducting first article tests, the sample shall be inspected by the contractor for all the requirements of the contract, all of the examinations and tests specified in Table 1 (Functional Test) and Table 11 (Special Sampling) herein and to all requirements of the applicable drawings. The sample and a record of this inspection, including CTRKOC, shall be submitted for the Government approval. The Government reserves the right to witness the Contractorts inspection.

4.3.2 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. units of product selected for inspection-shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity sha~l be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine-conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be

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performed by the supplier after first article approval and prior to **Government acceptance or rejection by item or lot.**

4.4.2.3 Functional tests. The tests in Table I shall be **performed on a 100% basis.**

TABLE I. Functional tests (100% Inspection)

CHARACTERISTIC		**REQUIREMENT**	TEST PROCEDURE
Critical:		**None**	N/A
Major			
101	Electrical	**3*4**	4.7.2
102.	Performance	**3.5**	4.7.3
Minor:		**None**	N/A

4.5 Special sampling. One 33-volt regulator out of the first 50 produced and one out of each **100 produc=d thereafter shall be** selected at random. The sample **shall have met the requirements and** tests of Table I and shall then **meet the tests in Table II. Unless otherwise specified in the contract the items shall be returned to the lot upon successful completion of testing.**

TABLE II. Special sampling

CHARACTERISTIC		REQUIREMENT	**TEST PROCEDURE**
Critical:		None	**N/A**
Major:			
103.	Vibration	3.3.1	4.7.1.1
104.	Shock	3.3.2	4.7.1.2
105.	Altitude	3.3.3	4.7.1.3
106.	Temperature	3.3.4	4.7.1.4
107.	Humidity	3.3.5	4.7.1.5
Minor:		None	N /a

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government **until** necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. . where the **contract provides for Government-furnished test equipment, supply**

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and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error In the test equipment design must be considered as part of the prescribed product tolerance limit. Thus , concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall resubmitted for approval in accordance with 6.3.

4.6.2.3 Test equi pment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table III shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE 111. Test equipment list

ITEM NUMBER	DESCRIPTION
1. Signal generator	Capable of producing a 22 ± 1 KHz, 71.5 ± 12.5 V peak-to-peak square wave. Output impedance less than 20 ohms.
2. Power supply	Dc , 36 to 53 V, 2 A, regulated to ± 0.5%.
3. Voltmeter	Dc, O to 60 V, ± 0.01% accuracy, 20,000 Ohmslvolt.
4. Oscilloscope	Input sensitivity: 20 mV/cm. Input impedance: 1 megohm. - Vertical passband: dc to 1 MHz. Horizontal deflection: 5us/cm

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5. Ammeter	Dc,	0 to 2A, 1% accuracy.
6. Resistive load		Variable 0 to 50 ohms, 100 W.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514.3, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph 11-2.1 apply. The 33-volt regulator shall be attached to the vibration generator directlY or with a fixture and securely held by its normal means of attachment. specified in vibration, the 33-volt regulator shall be tested as 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be perfomed as specified in Procedure X, section II of Method 516. 3 of MIL-STD-810 except that:

The post test functional check (Step 5 of procedure I need not be conducted until completion of all shock Impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and II ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frefquency response within ± 10% from 20 to 2,ooO hertz.

After the application of all the shock impulses, the 33-volt regulator shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures 11 and 111, Method 500.2 of MIL-STD-810. The operational checkout shall be as specified in 4-7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in procedures I and II, Method,501:2 of MIL-STD-810 with constant temperature extremes as, specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for complance with 3.3.4.

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4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507.2 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.7.3 Performance. The selected 33-volt regulators shall be tested as follows:

4.7.3.1 output terminals E71-E72. The following tests shall be performed with test equipment connected to the 33-volt regulator as shown in Figure 1.

4.7.3.1.1 Output voltage level potentiometer R2. Using signal generator (Table III, item 1), apply a 22 ± 1 kHz, 71.5 ± 12.5 volt peak-to-peak square wave to P1-1 and P1-2. Using power supply (Table III, item 2), apply 44.5 ± 8.5 VdC between P1-6 (+) and P1-7 (-). with switch S1 open and using voltmeter (Table III, item 3), verify that the voltage between E71 (+) and E72 (-) can be adjusted to 33.00 +_0,10 Vdc with R2,

4.7.3.1.2 Current limit potentiometer R17. Perform the following steps in sequence:

a. Set up conditions specified in 4.7.3.1.1.

b. Adjust R2 for a voltage indication of 33.00 ± 0.10 v.

c. Adjust R17 fully CCW.

d. Close load switch S1 (See Figure 1).

e. Close short switch S2 (See Figure 1).

f. Adjust R17 CW until the output current equals 1.6 Adc .

g. Open S2.

h, Vary the resistance load {Table III, item, 6) to obtain a current indication of 0.75 Adc.

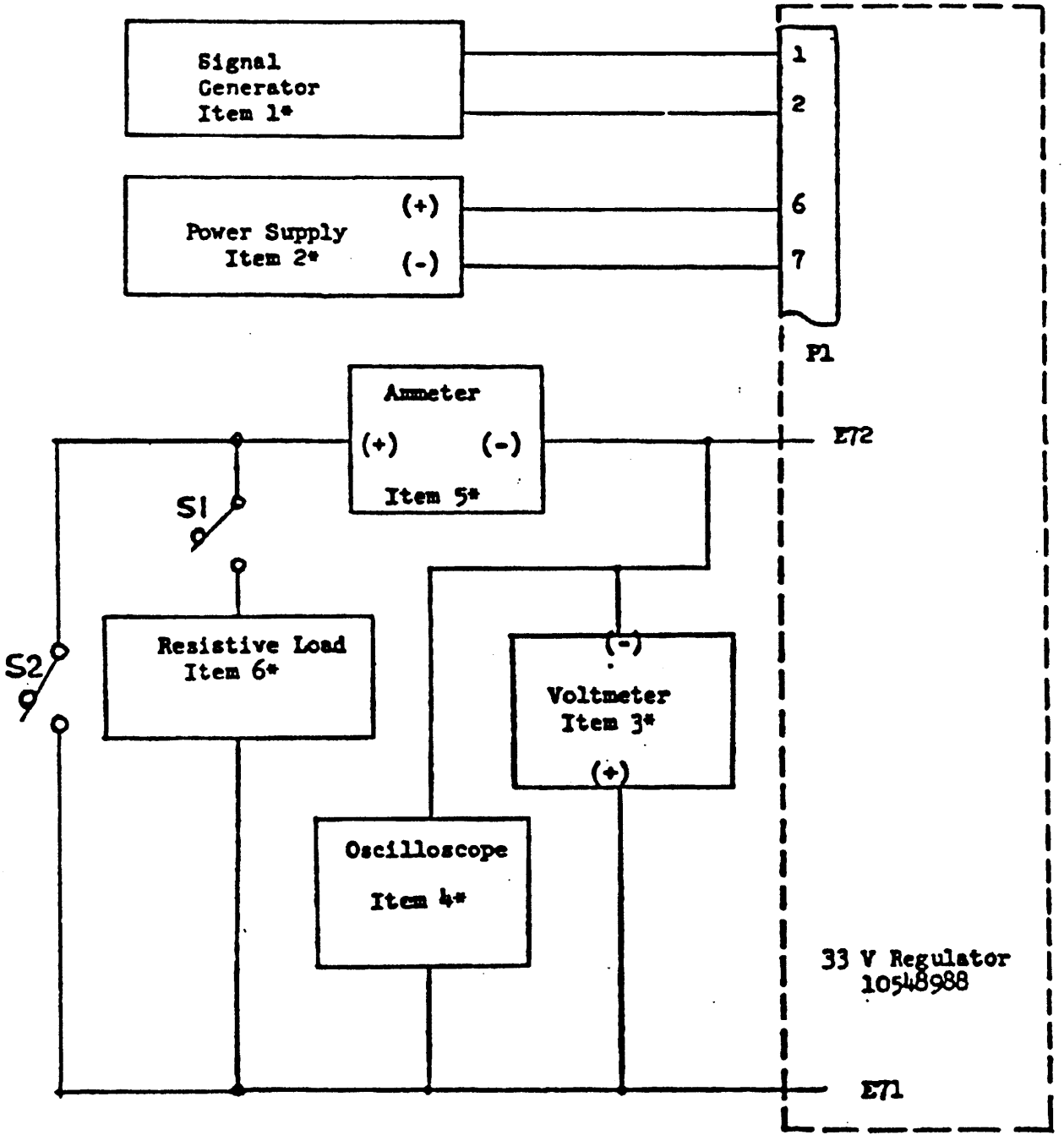
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i. **Record the voltage indication using voltmeter (Table III,** item 3).

j. **Close s2 momentarily, then** open.

k. **Using voltmeter (Table III, item 3), verify the output** automatically returns to within ± 0.10 **volts** of the level in "i"above.

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*Refer to 4.6.2.3

FIGURE 1. 33-Volt regulator test setup for E71 and E72.

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4.7.3.1.3 Voltage regulation, line. With R2 and R17 adjusted as specified in 4.7.3.1.2, adjust the resistive load (Table III, item 6), for a current indication of 0.75 Adc. Vary input dc supplied by power supply (Table III, item 2) from 36 to 53 Vdc. Using voltmeter (Table III, item 3), verify that the output voltage does not vary more than ± 0.33 Vdc.

4.7.3.1.4 Voltage regulation, load. With R2 and R17 adjusted as specified in 4.7.3.1.2, set the input voltage supplied by power supply (Table III, item 2) to 44.5 ± 8.5 Vdc. **Adjust resistive load (Table 111, item 6) for maximum resistance and open switch S1. Record the voltage indication using voltmeter (Table III, item 3). Close switch S1 and vary the resistive load to obtain 0.75 Adc. Using voltmeter (Table III, item 3), verify that the output voltage - varies no more than ± 0.33 V.**

4.7.3.1.5 Output ripple. With input voltage supplied by power supply (Table III, item 2) set for 44.5 ± 0.44 Vdc and with R2 and R17 adjusted as specified in 4.7.3.1.2, adjust the resistive load for a current indication of 0.75 Adc. Using oscilloscope (Table III, item 4), verify the ripple does not exceed 0.33 V peak-to-peak excluding high-frequency spikes which occur at approximately 25 psec. intervals.

4.7.3.2 Output terminals E73-E74. The following tests shall be performed with test equipment connected to the 33-volt regulator as shown in Figure 2.

4.7.3.2.1 Output voltage level potentiometer R25. Using signal generator (Table 111, **item 1), apply a 22 ~ 1 KHz, 71.5 ~** 12.5 V, peak-to-peak square wave to P1-3 and P1-4. Using power supply {Table III, item 2), apply 44.5 ± 8.5 Vdc between PI-8 (+) and P1-9 (-). With switch S1 open and using voltmeter (Table 111, item 3), verify the voltage between E73 (+) and E74 (-) can be adjusted to 33.00 ± 0.10 Vdc with R25.

4.7.3.2.2 Current limit potentiometer R40. Perform the following steps in sequence:

a. **Set** up conditions specified in 4.7.3.2.1.

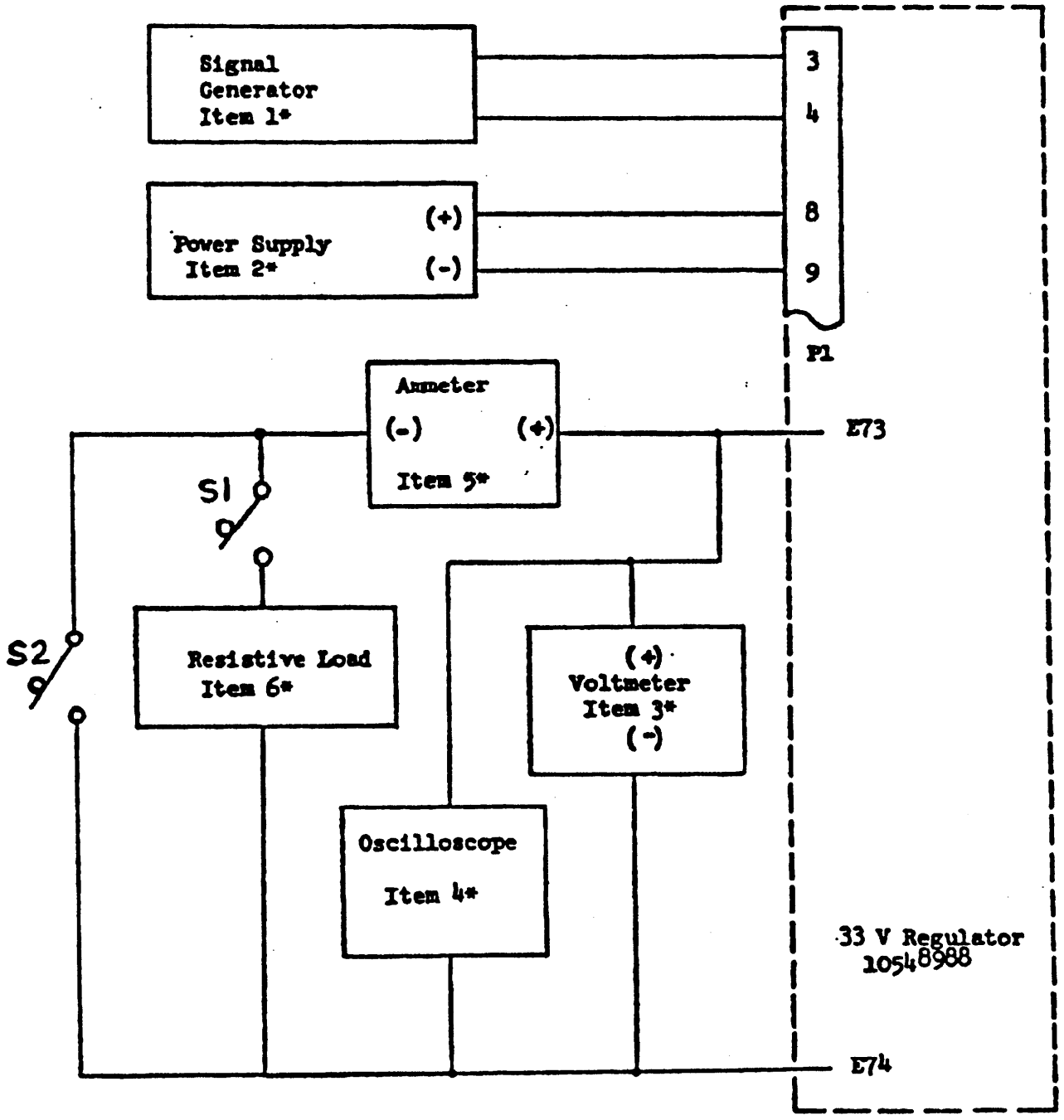
b. **Adjust R25 for a voltage indication of 33.00 ± 0.10 vdc.**

c. Adjust R40 fully CCW.

d. **Close** load **switch S1 (See Figure 2).**

e. Close short switch S2 (See Figure 2).

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*Refer to 4.6.2.3

FIGURE 2. 23-Volt regulator test setup for E73 and E74.

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f. Adjust R40 clockwise until the output current equals 1.6 Adc.

g. Open S2.

h. Vary the resistive load (Table III, item 6) to obtain a current indication of 0.75 Adc.

i. Record the voltage indication using voltmeter (Table 111, item 3).

j. Close S2 momentarily, then open.

k. Using voltmeter **(Table 111, item 3), verify the** output automatically returns to within ± **0.10 volts** of the level in "i" above.

4.7.3.2.3 Voltage regulation, line. **With R25 and R40 adjusted as specified in 4.7.3.2.2, adjust resistive load (Table 111, jtem 6) for a current indication of 0.75** Adc. Vary the input dc supplied by power supply (Table III, item 2) from 36 to 53 Vdc. Using **voltmeter (Table III, item 3), verify that the output voltage** does not vary more than ± **0.33 Vdc.**

4.7.3.2.4 Voltage regulation, load. With R25 and R40 adjusted as specified in 4.7.3.2.2, set the input voltage supplied by power supply (Table III, item 2) to 44.5 ± 8.5 Vdc. Adjust resistive load (Table III, item 6) for maximum resistance and open switch S1. Record the voltage indication using voltmeter (Table III, item 3). Close switch S1 and vary the resistive load **(Table III, item 6) to obtain 0.75 Adc. Using voltmeter (Table III, item 3), verify that the output** voltage does not vary more **than ± 0.33 V.**

4.7.3.2.5 Output ripple. With the input voltage supplied by power supply (Table 111, item 2) set for 44.5 ± 0.44 Vdc and with R25 and R40 adjusted as specified in 4.7.3.2.2, adjust the resistive load (Table 111, item 6) for a current indication of 0.75 Adc . **Using oscilloscope (Table III, item 4), verify that the ripple does not exceed 0.33 V peak-to-peak excluding high-frequency spikes which occur at approximately 25 psec. intervals.**

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in **accordance with Packaging Data Sheet PS10548988. The level of protection shall be as specified in the procurement document.**

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 33-volt regulator is used in Power supply Subassembly 10549013, a part of Power Supply PP4812/VPS-2, part of Radar Set AN/VPS-2. The 33-volt regulator is also used in the PP6324/TPM-22, part of Test Set AN/TPM-22. The 33-volt regulator provides the necessary voltages for functional operation of Radar Set AN/vPS-2 and provides the necessary test voltages in the AN/TPM-22 for functional checkout of Radar Set AN/VPS-2.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

5 **Title,** number and date of this specification.
Applicable stock number.
c. First Article Sample requirements.
d. Packaging requirements, if other than specified in Section 5.
em Serialization requirements, if applicable.
f. Certificate of conformance for each lot of shipment of product.

6.3 Submission of **contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I , Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.**

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared **by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC .**

6.5 Subject term **(keY word) listing.**

Fire control System
Power Supply Subassembly PP4812/VPS-2
Power Supply Subassembly PP6324/TPM22
Radar Set AN/vPS-2
Radar Test Set AN/TPM-22
Vads
Voitage regulator

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6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to **the** previous issue due to the **extensiveness of the changes.**

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A708)

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-50327C (AR)	30 APRIL 1990

3. DOCUMENT TITLE: REGULATOR, VOLTAGE: 10548988

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION & SPECIFICATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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